COLLEGE OF
INNOVATION & TECHNOLOGY

Assignment 1

Deadline: Sunday 09/07/2025 @ 23:59

[Total Mark for this Assignment is 100]

Student Details:

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CRN: CSC-575-01

Instructions:

- You must submit two separate copies **(one Word file and one PDF file)** using the Assignment Template on system via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template.
- You MUST show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- The work should be your own.
- Use **Times New Roman** font for all your answers.

Question

100 Marks

Write java program that:

- Ask the user to give a positive number between 10 and 20.
- Create an array of integer. The length of the array is the given number by the user.
- Print the content of the array
- Fill the array with numbers from 0 to the given number -1.
- Print the content of the array
- Fill the array with positive random numbers less than 200.
- Print the content of the array.
- Find and print :
- The odd numbers, the maximum and the minimum numbers.
- The sum and the average.

Typical output

```
Please give a number between 10 and 20

12

The state of the array before initialization

0  0  0  0  0  0  0  0  0  0  0  0

The state of the array after initialization

0  1  2  3  4  5  6  7  8  9  10  11

The state of the array after random filling

54  48  102  149  102  174  111  99  85  189  35  133

The odd numbers are

149 111 99 85 189 35 133

the sum is :1281.0

the avg is :106.75

the max is :189
```

the min is :35

AssignmentOne.java

100 Marks

```
//Import Scanner module
import java.util.Scanner;
//Import Arrays module
import java.util.Arrays;
//Import Random Module
import java.util.Random;
//Import Arraylist module
import java.util.ArrayList;

public class AssignmentOne {
    public static void main(String[] args) {
        //Ask for user input of a positive number between 10 and 20
        //Create an array of length equal to the number given by the user filled with zeroes
        int[] initialArray = initalArray(userInput());

        //Print the content of the array
        System.out.println("The state of the array before initialization:");
        System.out.println(Arrays.toString(initialArray));

        //Fill the array with numbers from 0 -> (user input - 1)
        int[] filledArray = fillArray(initialArray);

        //print the content of the array
        System.out.println("The state of the array after initialization:");
        System.out.println(Arrays.toString(filledArray));

        //Fill array with positive random numbers less than 200
```

```
        System.out.println("The state of the array after random filling:");
        int[] randomArray = randomArray(filledArray);

        //Print the content of the array
        System.out.println(Arrays.toString(randomArray));

         //Find and print: the odd numbers, the maximum value, the minimum value, the
sum, and the average
        arrayInfo(randomArray);
    }

    /**
     * Fucntion name: userInput
     *
     * @return (integer)
     *
     * Inside the function:
     * 1. Takes int from user
     * 2. Checks if input is int
     * 3. checks if number is between 10 and 20
     * 4. recursive call if entry is out of bounds
     *
     */
    public static int userInput() {
        Scanner scanner = new Scanner(System.in);
        System.out.println("Please give a number between 10 and 20");

        //Loop to handle all conditions
        if (scanner.hasNextInt()){
```

```
        int userNumber = scanner.nextInt();

        //check to see if number is between 10 and 20
        if (userNumber > 10 && userNumber < 20){
            scanner.close();
            return userNumber;
        }
        //if it isn't have them input another number
        else {
            System.out.println("That number was not between 10 and 20.");
            return userInput();
        }
    //if the input isn't a number have them try again
    }else{
        System.out.println("Please input and integer");
        return userInput();
    }
}

/**
 * Function name: inital Array
 *
 * @param value
 * @return
 *
 * Inside the function:
 * 1. creates a new array equal to the length of the value input into the function
 *
 */
```

```
public static int[] initalArray(int value){
    int[] newArray = new int[value];
    return newArray;
}
/**
 * Function name: fillArray
 *
 * @param array
 * @return
 *
 * Inside the function:
 * 1.loops through function with counter starting at 0 and assigns the index value to the coutner, then increments counter+1
 *
 */
public static int[] fillArray(int[] array){
    for (int i=0; i < array.length; i++){
        array[i] = i;
    }
    return array;
}
/**
 * Function name: randomArray
 *
 * @param array
 * @return
 *
```

```
 * 1.Loops through array and at each index assigns a random number between 0 and 200
 *
 */
public static int[] randomArray(int[] array){
    Random random = new Random();
    for (int i=0; i < array.length; i++){
        array[i] = random.nextInt(200);
    }
    return array;
}
/**
 * Function name: arrayInfo
 *
 * @param array
 *
 * Inside the function:
 * 1. creates an empty arraylist of odd numbers so we can easily add numbers from input array that are odd
 * 2. loop through array and add to odd Array if they are odd
 * 3. prints odd array
 * 4. loops through each index add them to total variable and prints sum
 * 5. calculates average and prints
 * 6. loop to find max value then prints
 * 7. loop to find min value then prints
 *
 */
public static void arrayInfo(int[] array){
    ArrayList<Integer> oddArray = new ArrayList<Integer>();
```

```
//loop through array and add to odd Array if they are odd
for (int i=0; i < array.length; i++){
    if ((array[i] % 2) > 0){
        oddArray.add(array[i]);
    }
}
//print result of oddArray
System.out.println("The Odd numbers in the array are: ");
System.out.println(oddArray);

//loop for sum
int sumTotal = 0;
for (int i=0; i < array.length; i++){
    sumTotal+=array[i];
}
//print total
System.out.println("The sum is: " + sumTotal);

//find average and print
double average = (double) sumTotal / array.length;
System.out.println("The average is: " + average);

//find and print out the max value of the array
int max_val = 0;
for (int i = 0; i<array.length; i++){
    if (array[i] > max_val){
        max_val = array[i];
    }
}
```

```
        System.out.println("The maximum value is: " + max_val);



        //find and print out the min value of the array
        int min_val = 200;
        for (int i = 0; i<array.length; i++){
            if (array[i] < min_val){
                min_val = array[i];
            }
        }
        System.out.println("The minimum value is: " + min_val);
    }
}
```

AssignmentOne.java Output

100 Marks

```
PROBLEMS 3   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\bradm\OneDrive\Desktop\Data Structures\Assignments>  & 'C:\Program Files\Amazon Corretto\jdk17.0.16_8\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessag
es' '-cp' 'C:\Users\bradm\AppData\Roaming\Code\User\workspaceStorage\a88551a9329dbf5fb5de85ecd637539a\redhat.java\jdt_ws\Assignments_ff83cc3\bin' 'AssignmentOne'
Please give a number between 10 and 20
h
Please input and integer
Please give a number between 10 and 20
50
That number was not between 10 and 20.
Please give a number between 10 and 20
15
The state of the array before initialization:
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
The state of the array after initialization:
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14]
The state of the array after random filling:
[169, 23, 19, 162, 150, 170, 138, 182, 170, 10, 77, 123, 22, 78, 170]
The Odd numbers in the array are:
[169, 23, 19, 77, 123]
The sum is: 1663
The average is: 110.86666666666666
The maximum value is: 182
The minimum value is: 10
PS C:\Users\bradm\OneDrive\Desktop\Data Structures\Assignments>
```

Please give a number between 10 and 20

h

Please input and integer

Please give a number between 10 and 20

50

That number was not between 10 and 20.

Please give a number between 10 and 20

15

The state of the array before initialization:

[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]

The state of the array after initialization:

[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14]

The state of the array after random filling:

[169, 23, 19, 162, 150, 170, 138, 182, 170, 10, 77, 123, 22, 78, 170]

The Odd numbers in the array are:

[169, 23, 19, 77, 123]

The sum is: 1663

The average is: 110.86666666666666

```
The maximum value is: 182
The minimum value is: 10
PS C:\Users\bradm\OneDrive\Desktop\Data Structures\Assignments>
```